

DA-9805 protects dopaminergic neurons from endoplasmic reticulum stress and inflammation

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ABSTRACT

Parkinson's disease (PD) is a multifactorial neurodegenerative disease with damages to mitochondria and endoplasmic reticulum (ER), followed by neuroinflammation. We previously reported that a triple herbal extract DA-9805 in experimental PD toxin-models had neuroprotective effects by alleviating mitochondrial damage and oxidative stress. In the present study, we investigated whether DA-9805 could suppress ER stress and neuroinflammation in vitro and/or in vivo. Pre-treatment with DA-9805 (1 µg/ml) attenuated upregulation of glucose-regulated protein 78 (GRP78), C/EBP homologous protein (CHOP) and cleaved caspase-3 in SH-SY5Y neuroblastoma cells treated with thapsigargin (1 µg/ml) or tunicamycin (2 µg/ml). In addition, DA-9805 prevented the production of IL-1β, IL-6, TNF-α and nitric oxide through inhibition of NF-κB activation in BV2 microglial cells stimulated with lipopolysaccharides (LPS). Intraperitoneal injection of LPS (10 mg/kg) into mice can induce acute neuroinflammation and dopaminergic neuronal cell death. Oral administration of DA-9805 (10 or 30 mg/kg/day for 3 days before LPS injection) prevented loss of dopaminergic neurons and activation of microglia and astrocytes in the substantia nigra in LPS-injected mouse models. Taken together, these results indicate that DA-9805 can effectively prevent ER stress and neuroinflammation, suggesting that DA-9805 is a multitargeting and disease-modifying therapeutic candidate for PD.

1. Introduction

Parkinson's disease (PD), one of the most common neurodegenerative diseases, is characterized by motor deficits including bradykinesia, rigidity, resting tremor, and loss of balance [1,2]. Its main pathological characteristic is the loss of dopaminergic neurons in the pars compacta

of the substantia nigra (SNc) and striatum (ST) [3]. Another important feature is the presence of intraneuronal proteinaceous inclusions (Lewy bodies) composed of fibrillary aggregates of misfolded α-synuclein (α-Syn) [4,5] that can activate microglia. Although molecular profiling of microglia in animal models of PD has not yet been reported, activated microglia are known to be prominent in mouse models and human PD

Abbreviations: AD, Alzheimer's disease; HD, Huntington's disease; PD, Parkinson's diseases; ER, endoplasmic reticulum; LPS, lipopolysaccharides; SNc, pars compacta of substantia nigra; SNr, pars reticulata of substantia nigra; ST, striatum; α-Syn, α-synuclein; LRRK2, leucine-rich repeat kinase 2; MPP⁺, 1-methyl-4-phenylpyridinium; MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; ERAD, ER-associated degradation; UPR, unfolded protein response; IRE1α, inositol-requiring enzyme 1 α; ATF-6, activating transcription factor-6; PERK, RNA-dependent protein kinase (PKR)-like ER kinase; XBP1, X-box binding protein-1; CHOP/GADD153, C/EBP homologous protein; Tg, thapsigargin; Tm, tunicamycin; SERCA, sarco/ER-associated Ca²⁺ ATPase; TNF-α, tumor necrosis factor-α; IL-1β, interleukin-1β; NO, nitric oxide; MCE, Moutan Cortex Radicis; BFE, *Bupleurum falcatum* L; ADE, *Angelica dahurica*; TMRE, tetramethylrhodamine ethyl ester; CM-H₂DCFDA, 5,6-chloromethyl-2',7'-dichlorodihydrofluorescein diacetate acetyl ester; DMEM, Dulbecco's Modified Eagle Medium; FBS, fetal bovine serum; DMSO, dimethyl sulfoxide; CMAPS, cell-based mitochondrial function analysis system; ROS, reactive oxygen species; OXPHOS, oxidative phosphorylation, MnSOD, Mn-superoxide dismutase; mtDNA, mitochondrial DNA; nDNA, nuclear DNA; NOx, nitrite/nitrate; HPMC, hydroxypropyl methylcellulose; i.p., intraperitoneal; PB, phosphate buffer; PBS, phosphate buffered saline; TH, tyrosine hydroxylase; Iba-1, ionized calcium binding adaptor molecule 1; GFAP, glial fibrillary acidic protein; ABC, avidin-biotin peroxidase complex; SEM, standard error of the mean; ANOVA, one-way analysis of variance; ΔΨ_m, mitochondrial membrane potential; iNOS, inducible nitric oxide synthase; 6-OHDA, 6-hydroxydopamine; Nrf2, nuclear factor erythroid 2-related factor 2.

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[6]. Despite extensive research, it remains unclear what causes the selective loss of dopaminergic neurons. Currently, there are no preventive or disease-modifying interventions due to a poor understanding of the molecular mechanism of the pathogenesis of PD.

Only ~5% of patients with PD have genetic mutations (familial PD). The rest, 95% of PD, are considered to have a sporadic component (sporadic PD). Many familial PD-linked genes are involved in mitochondrial function [1]. For example, leucine-rich repeat kinase 2 (LRRK2) mutations, as the most common cause of autosomal-dominant PD, can regulate mitochondrial dynamics [7] and endoplasmic reticulum (ER)-mitochondrial tethering [8]. Mitochondrial toxins such as 1-methyl-4-phenylpyridinium (MPP⁺) and rotenone can induce PD in experimental models that resemble sporadic PD. Furthermore, mitochondrial activation by mitochondrial modulators such as TFAM and NRF-1 could reverse MPP⁺-mediated nigrostriatal neurodegeneration [9]. These studies suggest that mitochondrial dysfunction might be a major cause of dopaminergic neurodegeneration.

ER stress is recognized as another important neuropathological feature of PD [10–12]. The accumulation of misfolded pathological proteins is commonly involved in all neurodegenerative diseases such as Alzheimer's disease (AD), Huntington's disease (HD), and PD [12]. Lewy bodies found in PD are believed to be ER-stress-induced aggregates of α -Syn, although their association with the PD phenotype is not straightforward [13]. Accumulation of unfolded and misfolded proteins in the ER can lead to accumulation of ER-associated degradation (ERAD) substrates and activate the unfolded protein response (UPR) signaling pathway to eliminate unfolded/misfolded proteins and maintain homeostasis [14]. UPR signaling is mediated by activation of three stress sensors located in the ER membrane, including inositol-requiring enzyme 1 α (IRE1 α), activating transcription factor-6 (ATF-6), and RNA-dependent protein kinase (PKR)-like ER kinase (PERK) [15,16]. In postmortem tissues from sporadic cases of human PD, signs of ER stress have been observed, including phosphorylation of PERK [17] and accumulation of ERAD substrates, LRRK2, and α -Syn [18]. Clear activation of UPR has also been reported in neurotoxin-based in vivo models of PD, where ATF6, X-box binding protein-1 (XBP1), and C/EBP homologous protein (CHOP/GADD153) play roles in controlling dopaminergic neuron survival [19,20]. In fact, ER stress induced by thapsigargin (Tg) or tunicamycin (Tm) can perturb mitochondrial function and the Akt signaling pathway, resulting in neuronal cell death [21]. Treatment with Tg, an inhibitor of sarco/ER-associated Ca²⁺ ATPase (SERCA), or Tm, an inhibitor of N-linked protein glycosylation, can prevent proper folding of ER-Golgi secretory proteins [11,22,23]. Furthermore, intranigral injection of Tm can induce ER stress and trigger α -Syn oligomerization, resulting in PD phenotypes in vivo [24]. However, the initial induction of ER stress is considered indirect in neurodegeneration, although protein misfolding is a common feature of neurodegenerative diseases. Nevertheless, the intervention of ER stress is still a good strategy for treating PD.

On the other hand, the neuroinflammatory process is also observed in postmortem and in vivo studies of PD [25]. Neuroinflammation may contribute to cascade events leading to dopaminergic neuronal degeneration. Its underlying mechanism comprises microglial activation, astrogliosis, and lymphocyte infiltration that are commonly found in other neurodegenerative disorders [26,27]. Several lines of evidence have suggested that neuroinflammation could be involved in the progression of neurodegeneration by producing deleterious proinflammatory cytokines and neurotoxic materials, including tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), IL-6, and nitric oxide (NO) [28,29]. In addition, it has been suggested that enhanced NF- κ B activation might be responsible for sustaining the activated state of pro-inflammatory microglia and progressive neurodegeneration in PD [6]. Therefore, therapeutic intervention against inflammatory responses has been suggested to help prevent disease progression or even stop the pathological process of PD [29]. All three cellular processes of mitochondrial dysfunction, ER stress, and neuroinflammation have been

observed in PD patients and animal models. Therefore, regardless of the cause of PD, targeting all of these processes involved in the progression of PD should be more beneficial as a therapeutic strategy to treat PD than targeting only one of them.

DA-9805 is a mixed herbal ethanol extract containing Moutan cortex Radicis (MCE, *Paeonia suffruticosa* Andrews) [30], *Bupleurum falcatum* L (BFE) [28], and *Angelica dahurica* (ADE) [31]. DA-9805 can recover mitochondrial activities [32] and neurotoxicity [33] in toxin-induced models of PD. In this study, we investigated whether DA-9805 could also target ER stress and neuroinflammation using an in vitro model of ER stress induced by Tg or Tm and an in vivo model of neuroinflammation induced by lipopolysaccharide (LPS). DA-9805 was an excellent candidate as a disease-modifying therapeutic drug against PD.

2. Materials and Methods

2.1. Materials

Lipopolysaccharide (LPS, 100 ng/ml, *Escherichia coli* (O111:B4)), 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazoliumbromide (MTT), and Griess reagent (sulfanilamide, N-(1-naphthyl)-ethylene-diamine dihydrochloride) were purchased from Sigma-Aldrich (St. Louis, MO, USA). LPS stock (100 μ g/ml) diluted in phosphate-buffered saline (PBS) was stored at -20 °C until use. Calcein AM, MitoSox, tetramethylrhodamine ethyl ester (TMRE), and 5,6-chloromethyl-2',7'-dichlorodihydrofluorescein diacetate acetyl ester (CM-H₂DCFDA) were purchased from Molecular Probes (Eugene, OR, USA). The ATP content Assay kit for (CellTiter-Glo) was purchased from Promega Co. (Madison, WI, USA). The media and culture reagents were products of Gibco Industries Inc. (Auckland, New Zealand). All other reagents were purchased from Sigma-Aldrich.

2.2. Preparation of DA-9805

A standardized DA-9805 was prepared by Dong-A ST (Yongin, South Korea) as described previously [32]. Briefly, dried and meshed plant material mixture of MC, BR, and ADR at a weight ratio of 1:1:1 was extracted with 90% ethanol on a stirring plate for 24 h at room temperature (15–25 °C), filtered, and evaporated under reduced pressure at 45–50 °C. The viscous ethanol extract of the mixture was freeze-dried to obtain a powdered extract and kept at 4 °C (yield: 6.7%) until use.

2.3. Cell culture, transfection, and treatment

SH-SY5Y human neuroblastoma cells (ATCC®, CRL-2266™; Manassas, VA, USA) were first cultured in Dulbecco's Modified Eagle Medium (DMEM)/F12 supplemented with 10% fetal bovine serum (FBS), 100 U/ml penicillin, and 100 μ g/ml streptomycin (complete media, CM) at 37 °C with 5% CO₂. These cells were cultured in 60-mm or 96-well plates until > 95% confluency, followed by incubation in serum deficient media (SDM, DMEM/F12 containing 0.5% FBS) for 16 h. Cells in SDM were pretreated with DA-9805 (1 μ g/ml) for 4 h, followed by incubation with MPP⁺ (1 mM), Tg (1 μ g/ml), Tm (2 μ g/ml), or dimethyl sulfoxide (DMSO) for 20 h and assayed as described previously [32]. Creatine (50 μ M), rasagiline (100 nM), and ropinirole (50 μ M) were used for treatment as reference controls.

BV2 murine microglia cells (ATCC, CRL-2469) were cultured in DMEM complete media. Stable pNF- κ B-luc-transfected BV2 cells were prepared as previously described [28]. BV2 cells (5 $\times$ 10⁴ cells/well) were cultured in confocal dishes (60-mm or 96-well plates) for 24 h followed by incubation in serum free media (SFM) for 16 h. Cells in SFM were pretreated with DA-9805 (0.5, 1, 3 μ g/ml) for 4 h, followed by incubation with LPS (100 ng/ml) or DMSO vehicle for 20 h. The final concentration of DMSO was 0.1%.

2.4. Mitochondrial activity assays

Based on previously described detection methods, a systemic cell-based mitochondrial function analysis system (CMAPS) was established using SH-SY5Y cells in 96-well plates [9,32]. The assay system covers quantitative assays for viability (Calcein), methyl thiazyl tetrazolium-mitochondrial dehydrogenase activity (MTT) assay for NADH dehydrogenase complex 1 activity, ATP content, TMRE-based mitochondrial membrane potential (TMRE), and CM-H₂DCFDA- or MitoSox-dependent reactive oxygen species (ROS) of cells. Functional damage to the mitochondria was evaluated using CMAPS results.

2.5. Real-time qRT-PCR

Total RNA was isolated with Trizol reagent (Invitrogen, Carlsbad, CA, USA). Total RNA (2 µg) was reverse transcribed using MMLV reverse transcriptase (Promega, Madison, WI, USA) and 10 pM oligo dT primers (Invitrogen) according to each manufacturer's instructions. Real-time quantitative RT-PCR (qRT-PCR) was performed using primers for the ER stress marker genes (glucose-regulated protein 78 (GRP78) and sXBP-1), mitochondrial DNA (mtDNA)-encoded oxidative phosphorylation (OXPHOS) subunits (ND4, ND6, COX2), nuclear DNA (nDNA)-encoded OXPHOS subunits (NDUFA6, COX7B, ATP6), and mitochondrial related factors (Mn-superoxide dismutase (MnSOD), cytochrome C (CYCS)) on Roter-Gene Q (Qiagen, Hilden, Germany) with 2x Ampi-Gene® qPCR green Mix Lo-ROX (Enzo Biochem, NY) at 95 °C for 10 min, followed by 45 cycles of 95 °C for 5 s, 60 °C for 15 s, and 72 °C for 20 s. Measurements were performed in duplicate for each sample. The quantity of mRNA was normalized by simultaneous measurement of nuclear DNA encoding 18 S rRNA. The relative gene expression was determined using the 2^{-ΔΔC_t} method [34]. Relative levels of mRNA expression are presented as fold changes compared to those under the control condition. The qRT-PCR primers used in this study are summarized in [Supplementary Table S1](#).

2.6. Western blot analysis

SH-SY5Y cells treated with Tg (1 µg/ml) or Tm (2 µg/ml) for 24 h were harvested and extracted with protein extraction solution (iNtRON Biotechnology, Korea). Protein extracts (20–30 µg) were separated by 12% SDS-PAGE and analyzed by Western blotting and an enhanced chemiluminescence system (ECL, Amersham bioscience, NJ, Korea). Primary antibodies against GRP78 and CHOP were purchased from Santa Cruz Biotech (Dallas, TX, USA). Antibodies against caspase-3 and cleaved caspase-3 and horseradish peroxidase (HRP)-conjugated secondary antibodies were purchased from Cell Signaling Technology (Beverly, MA, USA). The equivalent loading of the protein was verified using an anti-β-actin antibody (Santa Cruz). Band intensities were quantified by densitometry. Protein expression levels were normalized to those of β-actin and shown as bar graphs.

2.7. Griess assay for NO measurement

BV2 cells (1 × 10⁵ cells/well) in 96-well plates in SFM were pretreated with DA-9805 and stimulated with LPS as described above. The concentration of nitrite/nitrate (NO_x) released from activated BV2 cells in the culture medium was assessed by the Griess method (Promega, Madison, WI, USA). Culture medium (100 µl) was incubated with 100 µl Griess reagent (5% sulfanilamide in hydrochloric acid, 2% naphthylethylenediamine in H₂O) at room temperature for 30 min in the dark. The absorbance at 540 nm was read using an ELISA microplate reader (Versamax, Molecular Devices, CA, USA). Standard calibration curves were prepared using sodium nitrite as a standard.

2.8. Assays for NF-κB activation

NF-κB activation was determined by two different assays: NF-κB response element-dependent luciferase reporter assay and immunostaining of nuclear localization of NF-κB p65/RELA [28]. The luciferase activities of BV2 stable cells transfected with pNF-κB-luc were measured using the luciferase reporter assay system (Promega) and a luminometer (Lumat LB 9501/16, Berthold, PA, USA). Relative light units were normalized to the protein concentration determined by the BCA assay (Pierce, Rockford, IL, USA). Nuclear localization of NF-κB was determined by immunostaining for NF-κB p65/RELA in BV2 cells cultured on poly-L-lysine coated coverslips in 6-well plates. The treated cells were fixed in 4% paraformaldehyde for 20 min and permeabilized with 0.1% triton X-100 for 15 min at room temperature. Polyclonal rabbit antibodies against anti-NF-κB p65/RELA (sc-372, Santa Cruz) were applied for 24 h (1:250), followed by 1 h incubation with FITC-conjugated goat anti-rabbit IgG. After being washed with PBS, the coverslips were mounted with GEL/MOUNT (Biomedica, Foster City, CA, USA). Fluorescence was visualized under a confocal fluorescence microscope (Carl Zeiss, Göttingen, Germany).

2.9. Systemically LPS-injected mice and DA-9805 treatment

Systemically LPS-induced neuroinflammation models of mice were prepared as previously described [28,35,36]. Briefly, male C57/BL6 mice (8 weeks, 19–23 g, Daehan Biolink Co., Ltd, Eumseong, Korea) were divided into five groups (n = 5/group): PBS-injected vehicle control (CTL), PBS-injected DA-9805 (30 mg/kg/day) control, three LPS-injected groups pretreated with DA-9805 (0, 10 or 30 mg/kg/day). All animals had free access to food and water. Mice were orally administered with DA-9805 (0, 10 or 30 mg/kg/day) in 3% hydroxypropyl methylcellulose (HPMC) daily for 3 days and subsequently injected with LPS (10 mg/kg, *i.p.*, Sigma, St. Louis, MO, USA) or PBS. Doses of DA-9805 [32] and LPS [28] were used in the same range as previously described. Three hours after the injection of LPS or PBS, the mice were transcardially perfused and fixed. Brain sections were prepared for immunochemical staining [37]. Maintenance and treatment of the animals were carried out in accordance with the Principles of Laboratory Animal Care (NIH publication No. 85–23, revised 1985) and the Animal Care and Use Guidelines (KHSASP-20–163) of Kyung Hee University, Seoul, Korea.

2.10. Brain tissue preparation

Mice were anesthetized by *i.p.* injection of chloral hydrate (40 mg/kg), transcardially perfused with a saline solution containing 0.5% sodium nitrate and heparin (10 U/ml), and then fixed with 4% paraformaldehyde dissolved in 0.1 M phosphate buffer (PB, pH 7.2). Whole-brain tissues were dissected from the skull, postfixed overnight with 4% paraformaldehyde in 0.1 M PB at 4 °C, and then stored in 30% sucrose solution in 0.05 M PBS at 4 °C until they sank. The brains were frozen-sectioned on a Cryostat (Microsystems AG, Leica, Wetzlar, Germany) in 30 µm thick coronal sections and stored in a cryoprotectant (25% ethylene glycol, 25% glycerol, 0.2 M PB, and water) at 4 °C until use [33]. All sections were collected in six separate series and processed for immunostaining as previously described [32].

2.11. Immunohistochemical staining

Brain sections (30 µm in thickness) containing SN and ST were incubated with rabbit anti-tyrosine hydroxylase (TH, 1:2000; Pel-Freez Biologicals, Rogers, AR, USA) for dopaminergic neurons. Brain sections (30 µm thickness) containing SN or hippocampus were incubated with rabbit anti-glial fibrillary acidic protein (anti-GFAP, 1:5000; Neuromics, Edina, MN, USA) or rabbit anti-ionized calcium binding adaptor molecule 1 (anti-Iba-1, 1:1000; Wako, Osaka, Japan), respectively,

followed by staining with biotinylated anti-rabbit IgG and an avidin-biotin peroxidase complex (ABC) standard kit (Vector Laboratories, Burlingame, CA, USA). Signals were detected by incubating sections with 0.5 mg/ml 3,3'-diaminobenzidine (Sigma, St. Louis, MO, USA) in 0.1 M PBS containing 0.003% H₂O₂. To quantify GFAP- or Iba-1-positive cells, stained brain sections were imaged under a bright-field microscope (Olympus Optical, Tokyo, Japan). The results were quantified by counting the number of TH-immunopositive neurons in the SN in brain tissue Section (5 sections/series) at x100 magnification. The TH-immunopositivity in ST was measured by the optical density of TH-positive fibers at $\times 40$ magnification using ImageJ software (National Institutes of Health, Bethesda, MD, USA). For Nissl staining, brain sections were dehydrated in a desiccator for 1 h, stained with 0.01% cresyl violet (Sigma-Aldrich) in water for 20 min, then cover-slipped, and analyzed under a bright-field microscope. The Nissl-stained neurons in the entire SN area were counted using Image J software (size, 0-infinity; circularity, 0–1000). The proportion of TH-immunopositive cells relative to the Nissl-stained neurons in the SN were quantified.

2.12. Stereological cell counting

The unbiased stereological estimation of the total number of TH-immunopositive neurons in the SN was made using the optical fractionator method performed on an Olympus computer-assisted stereological toolbox system version 2.1.4 (Olympus) as previously described [37]. The sections used for counting covered the entire SN from the rostral tip of the pars compacta (SNc) back to the caudal end of the pars reticulata (SNr) (anterioposterior, -2.06 to -4.16 mm from the bregma). The SN was delineated on a x1.25 objective and generated counting grid of $150 \times 150 \mu\text{m}$. An unbiased counting frame of known area ($47.87 \times 36.19 \mu\text{m} = 1733 \mu\text{m}^2$) superimposed on the image was placed randomly on the first counting area and systemically moved through all the counting areas until the entire delineated area was sampled. The actual counting was performed using a x100 objective. The estimate of the total number of neurons was calculated according to the Optical Fractionator Equation.

2.13. Quantitation of activated microglia and astrocytes

To quantify resting and activated microglia and astrocyte in the SN and hippocampus, coronal brain sections ($30 \mu\text{m}$ thickness) were collected (5 sections/series) and labeled with anti-Iba-1 or anti-GFAP antibodies. For quantitative analysis, the sections labeled with Iba-1 or GFAP were digitized and manually counted within preselected fields ($500 \times 400 \mu\text{m}$) of the SN and hippocampus (2 fields/animal). Activated microglia and astrocytes were classified and counted according to their morphologies. Briefly, resting microglia and astrocytes displayed small compact somata that exhibited long, thin, and ramified processes. In contrast, activated microglia and astrocytes exhibited marked cellular hypertrophy and retraction of processes such that the length of the process was less than the diameter of the soma compartment. Cells were sampled only if the nucleus was visible within the sectional plane and if the cell profiles exhibited distinct delineated borders [38].

2.14. Statistical analysis

Data shown are expressed as mean $\pm$ standard error of the mean (SEM). Statistical significance was evaluated by one-way analysis of variance (ANOVA) with Tukey post-hoc testing analysis using InStat (GraphPad Software, San Diego, CA, USA). Values of $p < 0.05$ were considered statistically significant.

3. Results

3.1. DA-9805 decreases the levels of GRP78 and CHOP protein and mRNA

In UPR induced by ER stress, the 78 kDa glucose-regulated protein GRP78/BiP is up-regulated for protein quality control and functions as the major ER chaperon protein [39]. ER stress also induces CHOP, which is involved in producing cytokines induced by ER stress in macrophages. In addition, spliced X-box binding protein 1 (sXBP1) is generated through two UPR signaling pathways, IRE1 and ATF6, in response to ER stress.

To check whether DA-9805 alleviated ER stress, SH-SY5Y cells were pretreated with DA-9805 ($1 \mu\text{g/ml}$) for 4 h and then incubated with an ER stress inducer Tg ($1 \mu\text{g/ml}$) or Tm ($2 \mu\text{g/ml}$) for 20 h. The expression levels of GRP78/BiP, CHOP, and sXBP-1 were determined to monitor the degree of ER stress. Western blot analysis revealed that pretreated DA-9805 suppressed GRP78 and CHOP protein levels up-regulated by Tg (Fig. 1A) and Tm (Fig. 1D). Real-time qRT-PCR showed that DA-9805 also significantly decreased GRP78 and sXBP1 mRNA levels induced by Tg or Tm (Figs. 1B and 1E). Since ER stress-induced apoptosis depends on the activation of caspase-3 through caspase 12 [40], caspase-3 and cleaved caspase-3 were determined by Western blot. Both Tg and Tm increased caspase-3 cleavage from 35 kDa to 19 kDa and 17 kDa fragments (Figs. 1C and 1F). Treatment with DA-9805 attenuated the increase in cleaved caspase-3 induced by Tg or Tm, indicating the prevention of ER-stress-induced apoptosis.

3.2. DA-9805 suppresses ER stress-induced mitochondrial damage

ER stress induced by Tg or Tm can impair mitochondrial function and block the insulin signaling pathway in SH-SY5Y cells [21]. However, DA-9805 can reverse MPP⁺-treated mitochondrial defects [32]. Therefore, we investigated whether DA-9805 also protected against Tg- or Tm-induced mitochondrial damage. SH-SY5Y cells were pretreated with either DA-9805 ($1 \mu\text{g/ml}$) or a reference compound ($50 \mu\text{M}$ creatine, 100 nM rasagiline, or $50 \mu\text{M}$ ropinirole) for 4 h prior to MPP⁺, Tg-, or Tm-treatment. Treatment with MPP⁺ was used as a positive control of mitochondrial damage. DA-9805 increased cell viability (Calcein), NADH dehydrogenase complex 1 activity (MTT), and intracellular ATP content in Tg- or Tm-treated SH-SY5Y cells (Fig. 2A). In ER-stressed cells, DA-9805 restored TMRE-mediated mitochondrial membrane potential ($\Delta\psi\text{m}$) and ROS levels in mitochondria (MitoSox) and cells (DCF-DA) to control levels (Fig. 2B). On the other hand, the three reference compounds did not improve ER stress-mediated damage to MTT, ATP, or TMRE- $\Delta\psi\text{m}$, despite recovering MPP⁺-mediated mitochondrial damage (Fig. 2). These results suggest that DA-9805 improved mitochondrial damage caused by MPP⁺ and indirect mitochondrial damage induced by ER stress (Fig. 2).

3.3. DA-9805 attenuates the decrease in mitochondrial gene expression mediated by ER stress

Expression levels of some mitochondria-related genes are significantly reduced in the SN and ST of PD mice injected with 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) [32]. In cells treated with Tg- or Tm, real-time qRT-PCR was performed to determine the levels of mtDNA-mediated OXPHOS subunits (ND4, ND6, COX2, ATP6), nDNA-mediated OXPHOS subunits (NDUFA6, COX7B), and mitochondria-related factors (MnSOD, CYCS). It was found that the mRNA levels of all the mitochondrial genes determined, complex 1 (ND4, ND6, NDUFA6), complex 4 (COX2, COX7B), and complex 5 (ATP6) of the OXPHOS complex, MnSOD, and CYCS, were significantly suppressed by ER stress ($p < 0.01$ vs. CTL, Fig. 3). DA-9805 normalized mRNA levels excluding nDNA-encoded genes in Tg-treated cells (Fig. 3A), while it normalized all determined mRNA levels in Tm-treated

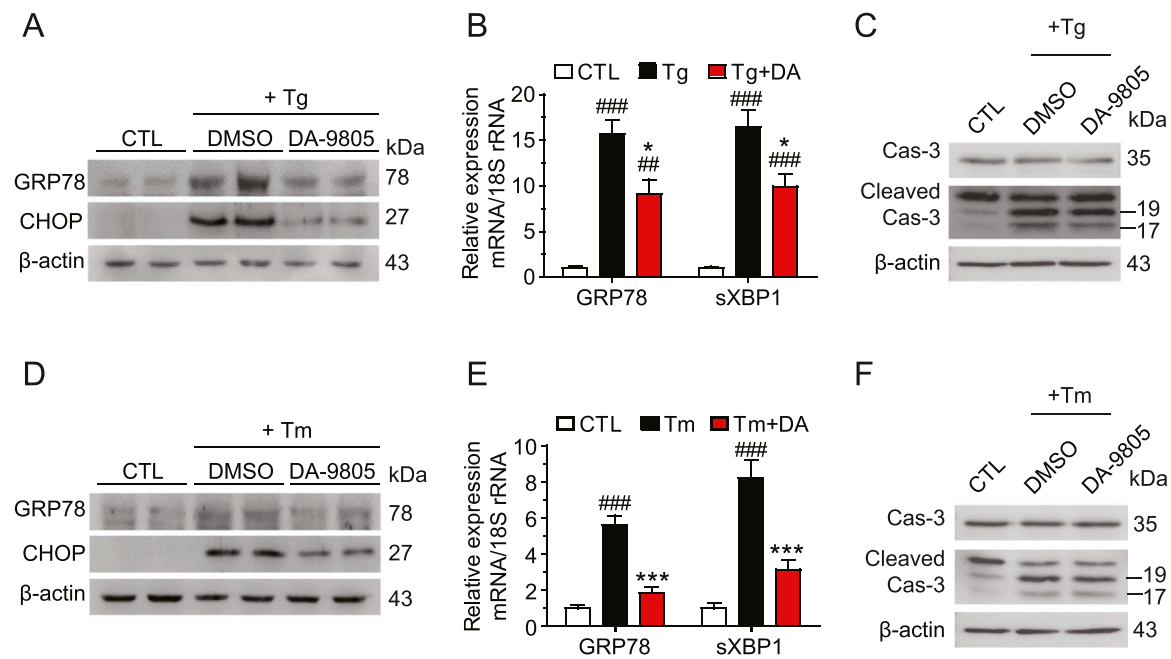


Fig. 1. DA-9805 suppresses Tg- or Tm-induced ER stress. SH-SY5Y cells in cell culture plates were pretreated with DA-9805 (1 μ g/ml) for 4 h and incubated with DMSO (CTL) or 1 μ g/ml Tg, 2 μ g/ml Tm for 20 h. (A, D) Western blot analysis of ER stress marker proteins (GRP78, CHOP). β -actin is used as a loading control. (B, E) Real-time qRT-PCR for determining mRNA expression levels of ER stress markers (GRP78, sXBP1). The mRNA level was normalized to 18 S rRNA. All values were given relative to DMSO control (CTL) with mean $\pm$ SEM (n = 6). Data represent the mean $\pm$ SEM. ### p < 0.01 and *** p < 0.001 vs. CTL; * p < 0.05; ** p < 0.01; *** p < 0.001 vs. Tg- or Tm-treated DMSO. (C, F) Western blot of caspase-3 (Cas-3, 35 kDa) and cleaved caspase-3 (19 kDa, 17 kDa).

cells (Fig. 3B).

3.4. Anti-neuroinflammatory effects of DA-9805 in BV2 cells

LPS-activated microglia cells can produce neuroinflammatory cytokines [41]. Upon stimulation by LPS in BV2 cells, mRNA expressions of IL-1 β , IL-6, and TNF- α were increased 1.5-fold, 66-fold, and 16-fold, respectively (Fig. 4A). DA-9805 decreased the LPS-induced increase of IL-1 β mRNA to normal levels and attenuated IL-6 and TNF- α mRNA levels by 40–50%. The release of NOx (nitrite/nitrate) into the culture medium by LPS-stimulated microglial cells was also significantly decreased by DA-9805 in a concentration-dependent manner (Fig. 4B). Furthermore, DA-9805 as low as 1 μ g/ml concentration normalized the level of NOx.

LPS stimulation induced phosphorylation and degradation of I κ B and consequently induced nuclear localization of active NF- κ B, a transcription factor of LPS-responsive cytokines or inducible NO synthase (iNOS) [28,42,43]. Immunofluorescence images of NF- κ B p65/RELA indicated that DA-9805 completely blocked LPS-mediated nuclear localization of NF- κ B p65/RELA (Fig. 4C). Furthermore, after examining the effect of DA-9805 on NF- κ B transcriptional activities, DA-9805 was found to reduce LPS-induced NF- κ B-dependent luciferase activities to normal levels in a dose-dependent manner (Fig. 4D). Taken together, these results suggest that DA-9805 can block nuclear translocation of NF- κ B and NF- κ B-dependent transcription of pro-inflammatory cytokines and iNOS in the nuclei of LPS-activated microglial cells.

3.5. DA-9805 inhibits the activation of astrocytes and microglia in the brains of LPS-injected mice

We have previously shown that systemic injection of LPS (10 mg/kg) can induce the activation of astrocytes and microglia and the loss of TH-positive neurons in the SN [28]. To examine the anti-neuroinflammatory effect of DA-9805, LPS (10 mg/kg, *i.p.*) was injected into mice pretreated with DA-9805 (0, 10, 30 mg/kg/day, *p.o.* 3 times) and brain tissues were harvested 3 h after LPS injection. The experiment scheme is shown in

Fig. 5A. SN (SNc, SNr) and ventral tegmental area (VTA) regions are marked in a representative TH-immunostained coronal section of the mouse midbrain (Fig. 5B). Coronal sections immunostained for TH demonstrated that acute neuroinflammation induced by a single injection of LPS decreased the total number of TH-positive cells in the SN and decreased TH-immunoreactive terminus in the ST (Fig. 5). Oral administration of DA-9805 protected such LPS-mediated decrease of TH-positive nigrostriatal dopaminergic neurons in the SN. The number of Nissl-stained neuronal cells in the SN was not significantly altered by any treatment. Fig. 5E showed the proportion of TH-immunopositive neurons relative to Nissl-stained neurons in the SN. The LPS-mediated reduction of the ratio of TH-positive cells was normalized by 30 mg/kg DA-9805. Treatment with DA-9805 (30 mg/kg) alone without LPS injection did not alter the number of TH-positive cells in the SN and TH-positive intensity of fibers in the ST.

When activated astrocytes or microglia were visualized by immunohistochemical staining of cryosections of brain tissues with GFAP (astrocytes) or Iba1 (microglia) antibodies, morphological activation characterized by enlarged cytoplasm, cell bodies with ramified processes, and intensified GFAP-positive and Iba1-positive staining was observed in LPS-injected brains (Fig. 6). DA-9805 inhibited LPS-induced neuroinflammation based on examinations for the activation of astrocytes and microglia. In DA-9805-treated SN, most GFAP-positive astrocytes and Iba1-positive microglia exhibited resting morphologies with small cell bodies and fewer branching processes (Fig. 6A, C). The count of activated astrocytes and microglia per section confirmed that DA-9805 suppressed the activation of both cells (Fig. 6B, D). Similar results were obtained for the hippocampus after immunochemical staining and counting of LPS-activated astrocytes and microglial cells (Supplemental Fig. S1). Treatment with DA-9805 significantly inhibited LPS-induced neuroinflammatory responses analyzed by the degree of activation of astrocytes and microglia in vivo. These data indicate that DA-9805 also has a therapeutic effect on neuroinflammation in vivo.

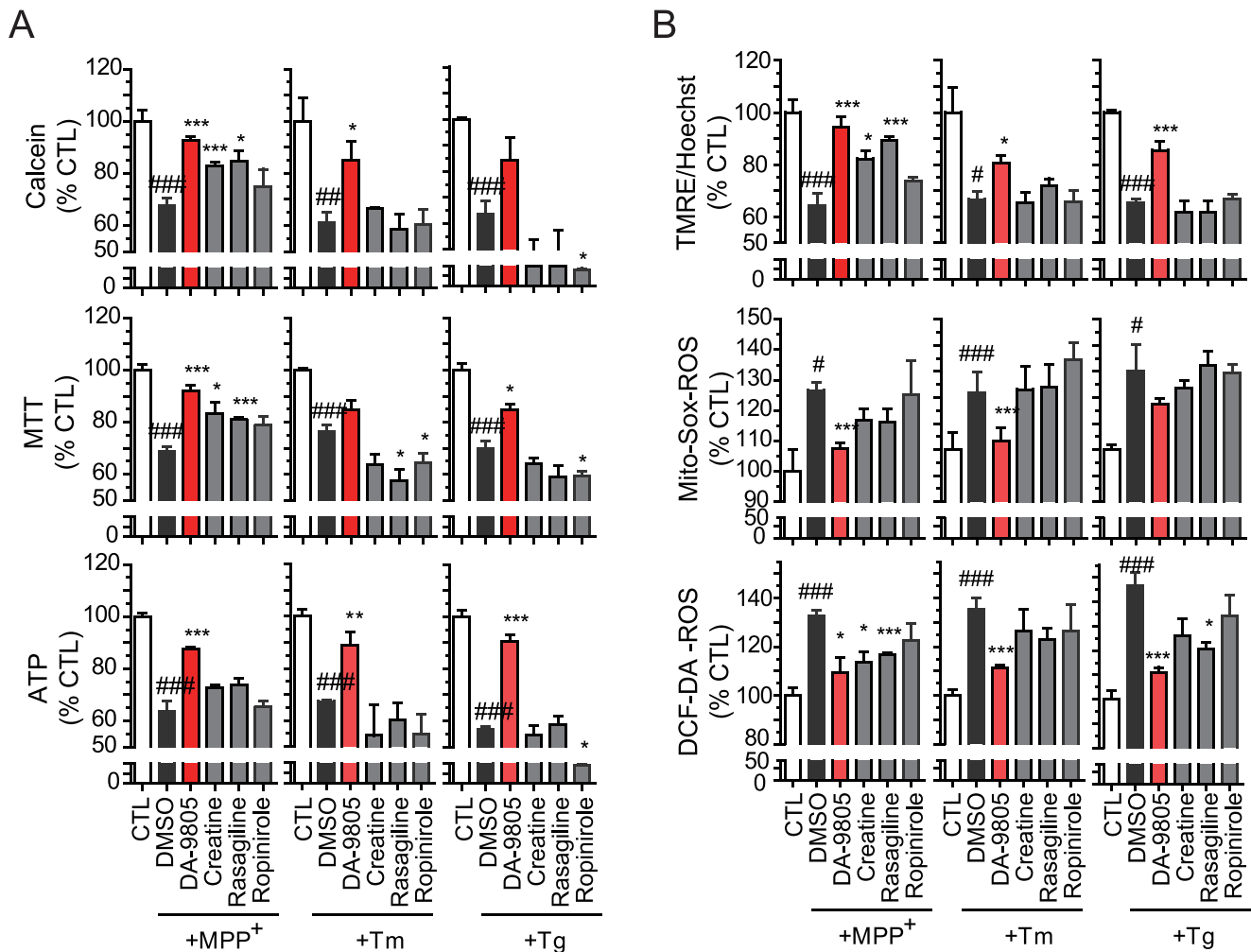


Fig. 2. DA-9805 protects from ER stress-induced mitochondrial damage in SH-SY5Y cells. SH-SY5Y cells were pretreated with DA-9805 (1 μ g/ml) or reference drugs (Creatine (10 μ M), Ropinirole (50 μ M), or Rasagiline (100 nM)) and then incubated with MPP⁺ (1 mM), Tm (2 μ g/ml), or Tg (1 μ g/ml) to induce mitochondrial damage. The harvested cells were assayed for (A) calcein, MTT, and ATP, (B) TMRE (membrane potential), MitoSox-ROS, and DCFDA-ROS. All values are given as % DMSO control (CTL) with mean $\pm$ SEM (n = 4). ## p < 0.01 and ### p < 0.001 vs. CTL; * p < 0.05; ** p < 0.01; *** p < 0.001 vs. DMSO treated with MPP⁺, Tg, or Tm.

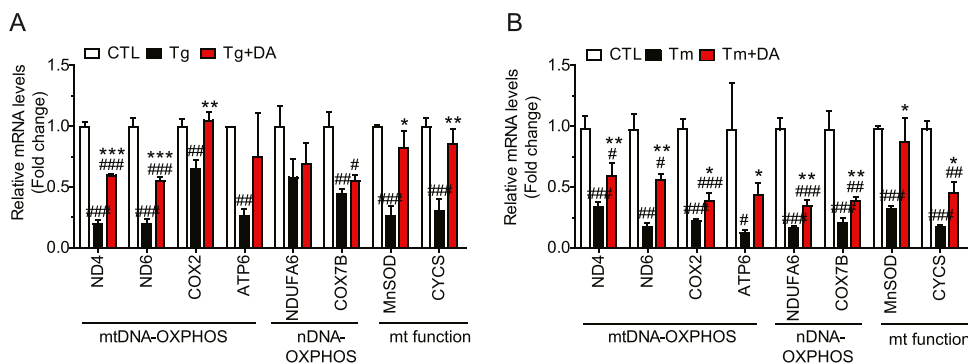


Fig. 3. Effects of DA-9805 on mitochondrial gene expression in Tg- or Tm-treated cells. SH-SY5Y cells were pretreated with DA-9805 (1 μ g/ml) for 4 h and incubated with (A) 1 μ g/ml Tg or (B) 2 μ g/ml Tm for 20 h. Real-time qRT-PCR was performed to determine the expression levels of mitochondrial genes, including mtDNA-mediated OXPHOS subunits (ND4, ND6, COX2, ATP6), nDNA-mediated OXPHOS subunits (NDUFA6, COX7B), and mitochondria-related factors (MnSOD, CYCS). The mRNA level was normalized to 18 S rRNA. All values are given as fold induction compared to DMSO control (CTL). They are presented as mean $\pm$ SEM (n = 6). Data represent the mean $\pm$ SEM. ## p < 0.01 and ### p < 0.001 vs. CTL; * p < 0.05; ** p < 0.01; *** p < 0.001 vs. DMSO.

treated with Tg or Tm.

4. Discussion

Previously, we have found that DA-9805 can attenuate MPTP- or 6-hydroxydopamine (6-OHDA)-induced dopaminergic neuronal cell death

by activating mitochondrial function [32] and anti-oxidative enzymes by stimulating the nuclear factor erythroid 2-related factor 2 (Nrf2) pathway [33]. The present study aimed to investigate the protective effect of DA-9805 against ER stress- and LPS-induced dopaminergic

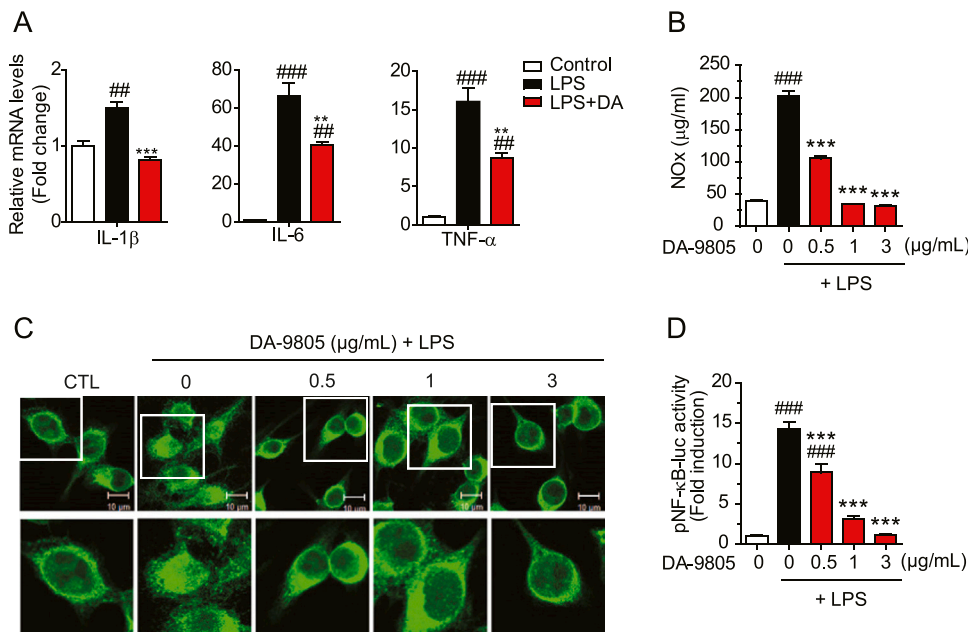


Fig. 4. DA-9805 suppresses neuroinflammation in LPS-stimulated BV2 cells. BV2 cells in SFM were pretreated with DMSO vehicle or DA-9805 (0.5, 1, 3 $\mu\text{g/ml}$) for 4 h and stimulated with LPS (100 ng/ml) for 20 h. (A) Real-time qRT-PCR results of IL-1 β , IL-6, and TNF- α mRNA expression in BV2 cells. DA-9805 (1 $\mu\text{g/ml}$) was used for pretreatment. (B) Quantification of total nitrite/nitrate (NOx). NOx concentrations in BV2 cell culture medium were determined by the Griess assay. (C) Confocal images of immunofluorescent staining of p65/RELA of BV2 cells. The lower images were enlarged from the white boxes of the upper images (magnification = 400x; scale bar = 10 μm). (D) pNF- κ B-luciferase activity. Stable BV2 cells transfected with NF- κ B-luc were pretreated with DA-9805 prior to stimulation with LPS (100 ng/ml). Luciferase activities were normalized to protein concentration. All data are given as fold induction compared to the DMSO control (CTL). They are expressed as mean $\pm$ SEM ($n = 6$). ## $p < 0.01$ and ### $p < 0.001$ vs. control; ** $p < 0.01$; *** $p < 0.001$ vs. DMSO treated with LPS.

neuronal disruption in vitro and in vivo. In SH-SY5Y neuroblastoma cells treated with Tg or Tm, pretreatment with DA-9805 (1 $\mu\text{g/ml}$) reduced the expression of ER stress markers, cleavage of caspase-3 (Fig. 1) and mitochondrial damage induced by ER stress (Fig. 2 and Fig. 3). In BV2 microglial cells, DA-9805 inhibited NF- κ B activation and blocked the production of LPS-induced IL-1 β , IL-6, and TNF- α (Fig. 4). Furthermore, oral administration of DA-9805 (10, 30 mg/kg) prevented LPS-induced activation of microglia and astrocytes and dopaminergic neuronal cell death in the SN (Figs. 5 and 6). Therefore, DA-9805 could relieve Tg- or Tm-induced ER stress and LPS-induced neuroinflammation, providing evidence that DA-9805 is indeed a promising multitargeting and disease-modifying therapeutic candidate for PD.

It has been suggested that PD could develop due to multiple factors, including mitochondrial defects, ER stress, and neuroinflammation, although its exact cause has not yet been understood [29]. To correct the dopaminergic deficit and thus alleviate motor symptoms in PD patients, current pharmacotherapeutics for PD are mainly dopamine replacement therapy, including levodopa, several dopamine agonists, inhibitors of monoamine oxidase (MAO) B and catechol-o-methyltransferase (COMT), and amantadine [44]. However, they do not directly target the disease mechanism. In addition, they have serious side effects after prolonged use. For example, long-term treatment with levodopa, the gold standard therapy for motor symptoms in PD, has led to motor complications such as involuntary movements (dyskinesias) [45]. Thus, the primary concerns of PD research have been the development of disease-modifying therapy that will replace symptomatic therapies.

To select disease-modifying therapeutics against PD, it is recommended to use animal models that exhibit most of the key features of PD, including the presence of Lewy bodies containing α -Syn and the loss of dopaminergic neurons leading to motor dysfunction known to be age-dependent and progressive. However, currently, there is no such perfect PD animal model [46,47]. Although acute neurotoxin (MPTP or 6-OHDA)-injected PD animal models can induce selective death of dopaminergic neurons and dopamine depletion, they do not reproduce the progressive nature of PD such as α -Syn aggregates. On the contrary, transgenic α -Syn mice show α -Syn aggregates with little death of dopaminergic neurons [48]. Similarly, there is no loss of dopaminergic neurons in mutant α -Syn or mutant LRRK2 transgenic mice [47]. When the virus-mediated mutant α -Syn is overexpressed or when preformed fibrils (PFFs) made of recombinant α -Syn proteins are inoculated [49], synucleinopathy is induced, but not PD-like behavior [50]. While these

animal models and cellular assays are useful for defining key pathways and neurotoxic factors for specific types of neurons, screening for multitargeting and disease-modifying therapeutics in the context of representing human disease is virtually very difficult. Instead, experimental models showing one or two PD phenotypes among dopaminergic neuronal death, synucleinopathy, mitochondrial dysfunction, ER stress, and neuroinflammation should be selected to demonstrate the specific disease-modifying effect.

TH-positive cell death was induced in two experimental models (in vitro ER stress and in vivo LPS-injected neuroinflammation) used in this study. It is not clear why LPS decreased the number of neurons immunoreactive to TH in the SN. Qin et al. [35] reported that a single systemic injection of LPS could induce a delayed and progressive loss of dopaminergic neurons in the SN, which began to appear 7–10 months after LPS injection in adult animals. It was suggested that the significant loss of TH-positive neurons might be due to neuronal death through peripheral inflammation induced by cytokines rather than down-regulation of TH expression. Dopaminergic neurons in the SN are more susceptible to inflammation compared to other regions of the brain. This is because the 4–5 times more microglia content in the SN may make the SN particularly susceptible to inflammatory insult. LPS can activate glial cells to release a wide range of proinflammatory and neurotoxic factors, cytokines, and lipid mediators [51]. Several studies to date have identified that cytokines like TNF- α and IL-1 β , iNOS-induced NO, and reactive oxygen species are main contributors to dopaminergic neuronal degeneration in LPS-induced PD models [35, 36]. On the contrary, other groups showed that LPS-injected mice did not induce a significant loss of TH immunoreactivity 2 weeks after treatment in the SN and ST [52,53]. However, no report demonstrates loss of TH-immunoreactivity or dopaminergic neuronal death 3 h after a single systemic injection of LPS, as in our study. To obtain acute neuroinflammatory model, we injected 10 mg/kg LPS into mice in this study and brain tissue was harvested 3 h after LPS injection. A recent RNA-seq transcriptome analysis study showed that brain astrocytes from LPS-stimulated mice rapidly expressed early responsive inflammatory genes, including master regulators of interferon-dependent transcription [54]. Expression levels of these early genes peak at 3 h after LPS injection and then decrease to normal levels at 72 h. Based on these observations, we propose that rapidly increased early inflammatory gene products in astrocytes might reduce TH-immunoreactivity in the SN and ST. More studies are required to elucidate the susceptibility mechanism

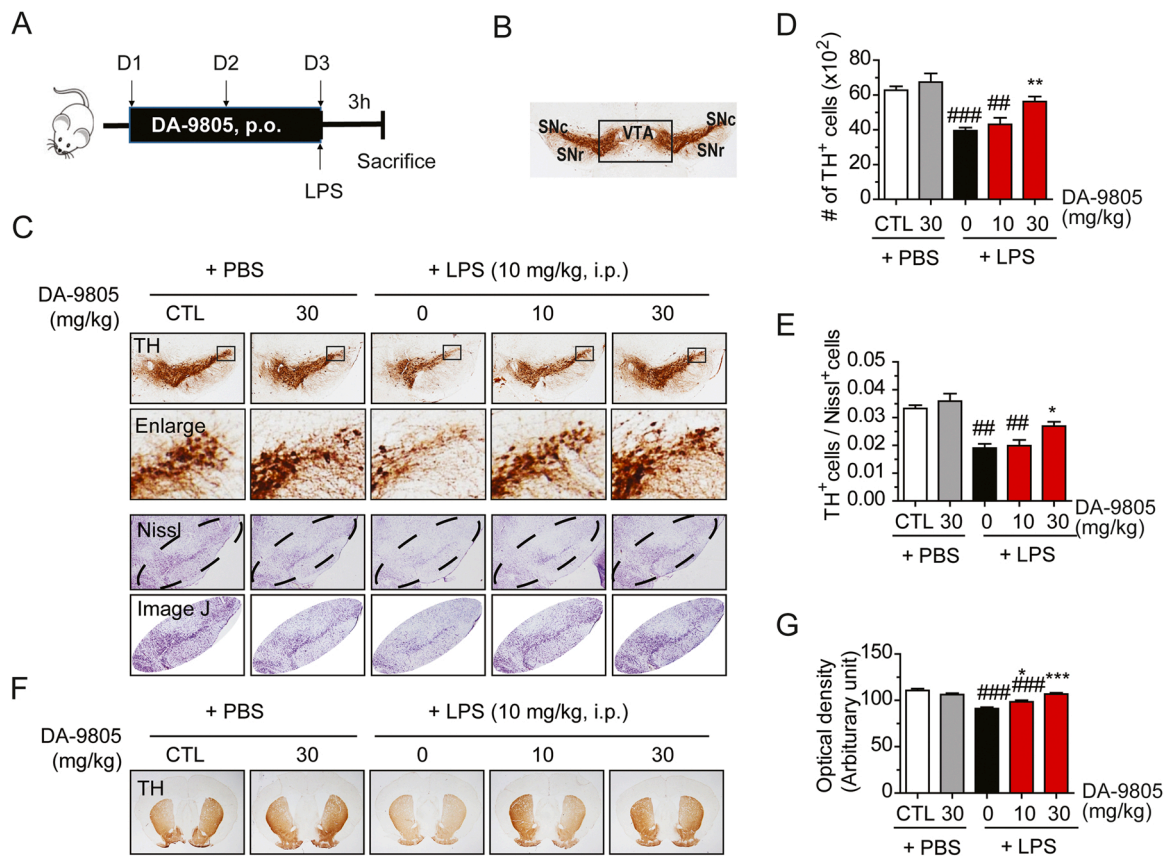


Fig. 5. Protective effects of DA-9805 on nigrostriatal dopaminergic neurons in LPS-injected mice. (A) Experimental scheme. Male C57/BL6 mice (8 weeks, 19–23 g) were divided into 5 groups ($n = 5$ mice per group): PBS+DA-9805 (0 (CTL), 30 mg/kg), LPS+DA-9805 (0, 10, 30 mg/kg). DA-9805 was orally administered once a day for 3 days before injection of LPS (10 mg/kg in PBS, i.p.). At 3 h after LPS injection, mice were sacrificed and transcardially fixed. The brain sections were then prepared for immunohistochemical staining. (B) A representative image of coronal midbrain sections containing the SNc, SNr and VTA immunostained for TH. Both sides corresponding to the SNc were used for stereological quantification of TH-positive neurons. (C) Representative images of coronal mesencephalic sections immunostained for TH (TH), the rectangle in the TH-stained image was enlarged (Enlarge), Nissl-stained neurons of the coronal section (Nissl, SNc area circumscribed by dashed line), and the SNc area where the number of Nissl-stained neurons was counted using Image J (Image J). (D) Quantification of the number of TH-positive neurons in the entire area corresponding to the SN in each section. Ten sections per animal were scored to quantify the total number of TH-positive cells in the SN. (E) Quantification of the ratio of the number of TH-positive neurons relative to the total number of Nissl-positive neurons in the SN. (F) Representative photomicrographs of coronal sections of the ST immunostained for TH. (G) Quantification of optical density of TH-positive fibers in ST. Data represent the mean $\pm$ SEM ($n = 5$). ^{##} $p < 0.01$ and ^{###} $p < 0.001$ vs. CTL; ^{*} $p < 0.05$, ^{**} $p < 0.01$ and ^{***} $p < 0.001$ vs. LPS-injected vehicle control.

of dopaminergic neurons to inflammatory stimulation. We obtained similar results for the hippocampus after immunochemical staining and counting of LPS-activated astrocytes and microglial cells (Supplemental Fig. S1). This is consistent with the report that LPS activates microglia leading to cognitive impairment and neuroinflammation in the hippocampus [55].

Pretreatment with DA-9805 effectively prevented ER stress, neuroinflammation, and interlinked mitochondrial dysfunction in both models. Meanwhile, as unpublished results, DA-9805 was found to have less effect on the dissolution of the already formed α -Syn aggregates, but had positive effects on reducing the formation of α -Syn aggregates induced by rotenone and reducing the synucleinopathy induced by PFF. These data suggest that DA-9805 can indeed reduce ER stress-mediated UPR in the PD model and may be effective in treating early stage PD patients. However, to prove the clinical applicability of DA-9805, future studies should test the simultaneous or posttreatment effect of DA-9805 with or after LPS (or other toxic factors).

Due to the complex nature of PD and the number of cell types involved, a disease-modifying therapeutic might need to contain more than one active component to confer a therapeutic benefit. To achieve this purpose, it would make sense to develop a blended extract of traditional herbal medicine that has been used safely for hundreds of years as neuroprotective agents, rather than combining several active

components [56]. DA-9805 was developed as a mixed herbal medicine consisting of MCE [30], BFE [28], and ADE [31] selected by screening for activities with anti-oxidation, anti-inflammation, or anti-ER stress among many traditional Korean herbal medicines [32,57]. The most abundant active components of DA-9805 are paeonol, paeoniflorin, imperatorin, and saikosaponins. They have been reported to prevent neurodegeneration caused by mitochondrial oxidative stress, ER stress, or neuroinflammation by independent groups [32,58]. Fig. 2 showed that DA-9805 attenuated most ER stress-induced mitochondrial damages, but not creatine, rasagiline and ropinirole, other known neuroprotective drugs used as reference. The causal factors of neurodegeneration are interconnected [21,59,60], showing that when one cause is controlled, other causes are alleviated accordingly (Figs. 2 and 3). Therefore, the effects of DA-9805 as a mixture of active components should be interpreted collectively rather than individually. Furthermore, to a similar extent as in the SN, oral administration of DA-9805 also protected TH-positive dopaminergic neurons in the VTA decreased by LPS (data not shown). These results suggest that it is necessary to consider multitargeting agents rather than single-target agents in developing neuroprotective agents for complex diseases like PD.

Despite the benefits of multitargeting, there are some difficulties in developing an extract containing one or more herbs as a disease-

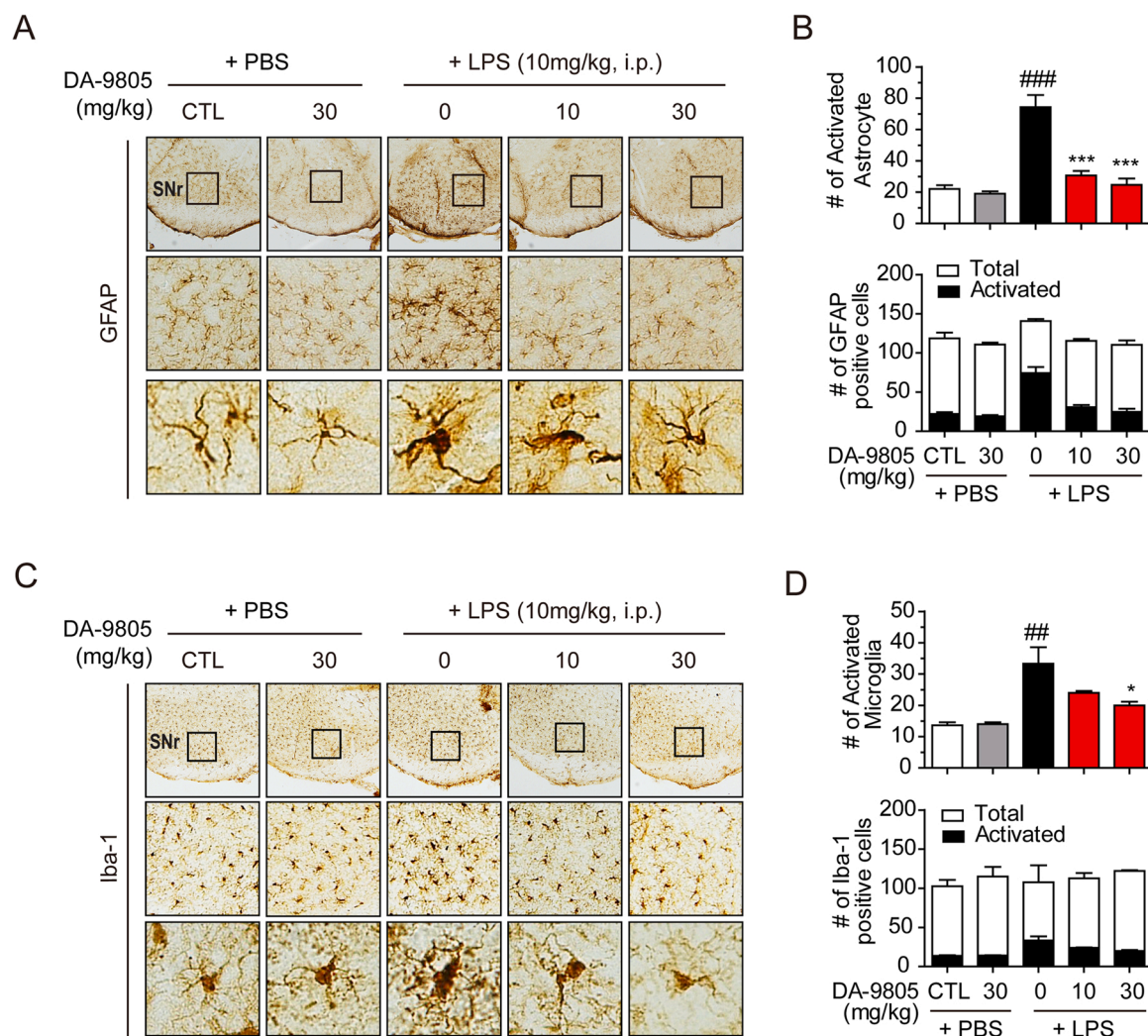


Fig. 6. DA-9805 suppresses LPS-induced activation of astrocytes and microglia in the SN. Mice brain sections were prepared as described in Fig. 5. (A, C) Representative images of coronal brain sections containing the SN immunostained with anti-GFAP (A, astrocytes) or anti-Iba-1 (C, microglia) antibodies. Box in SNr regions were enlarged to visualize activated astrocytes or microglia. (B, D) Quantitative analysis of astrocytes (B) or microglia (D). Upper graph, quantification of the number of activated astrocytes or microglia per section. Lower graph, comparison of the number of activated astrocytes or microglia (Activated) versus total GFAP- or Iba-1-positive cells (Total). Data represent the mean $\pm$ SEM ($n = 5$). $^{##}p < 0.01$ and $^{###}p < 0.001$ vs. CTL; $^{*}p < 0.05$; $^{***}p < 0.001$ vs. LPS-injected vehicle.

modifying medicine. First, it is hard to present a specific mode of action or drug target molecule, and the principle of drug development based on mechanism cannot be met. Second, the overall therapeutic effects of DA-9805 against PD cannot be comprehensively evaluated using current cell or animal models. Thus, DA-9805 should be applied to human PD patients with only fragmentary results obtained through PD models. In this case, the toxicity assessment of DA-9805 is critical. Since DA-9805 is composed of medicinal herbs that have been used for hundreds of years in Korea with little acute and chronic toxicity, it is expected that DA-9805 can suppress the progression of PD in humans without causing serious side effects after long-term use. Third, it may be difficult to stably secure three herbs that maintain the active component content of DA-9805 and establish a manufacturing process to prepare extracts with good quality control. We believe that pharmaceutical companies with extensive experience in the development and production of natural product medicines will overcome these shortcomings.

Mitochondrial dysfunction, ER stress, and neuroinflammation in the brain are prominent features of the most common neurodegenerative diseases such as AD, amyotrophic lateral sclerosis (ALS), HD, and PD [61]. Since DA-9805 can restore the common pathogenesis of these neurodegenerative disorders, DA-9805 could have great potential as a treatment for other neurodegenerative diseases associated with

mitochondrial dysfunction, ER stress, and neuroinflammation.

5. Conclusions

The current study shows that DA-9805 has neuroprotective effects against ER stress or LPS-induced neurotoxicity in *in vitro* and *in vivo* models. Furthermore, DA-9805 treatment can protect dopaminergic neurons and ameliorate mitochondrial dysfunction, ER stress, and neuroinflammation determined by their specific stress markers. Therefore, it can be concluded that DA-9805 as a mixture of herbal extracts might be an excellent candidate as a disease-modifying therapeutic against PD.

CCRediT authorship contribution statement

SK and YKP conceived the study and wrote the manuscript. SK, YP, YCK, and SL performed the experiments. SK and YKP analyzed the data and reviewed/edited the manuscript.

Conflict of interest statement

The authors declare no competing financial interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopha.2021.112389.

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